

theorem

Vera: Weakest Preconditions for TICL over Rust

From Imp Structural Lemmas to Rust VCGen Rules

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Background: TICL over Imp

TICL Structural Lemmas for Imp

Core idea (from the TICL paper)

For a small imperative language `StImp` with mutable state, the entailment $[t, m \Vdash_L \varphi]_S$ is proved structurally via inference rules on the syntax of t .

Key rules from `imp-lemmas.tex`:

- **Seq**_SAU_L: sequential composition decomposes into bind
- **If**_SAU_L: conditional branches both satisfy the obligation
- **While**_SAG: infinite loop with coinductive invariant $\mathcal{R}$
- **While**_SAU_L: terminating loop with ranking function f

Goal: Lift these to Rust syntax $\rightarrow$ define $\text{wp}(_, _)$ for Vera.

Notation

$\sigma \in \text{State}$ (mutable references, heap)

$[t, \sigma \Vdash_L \varphi]_{\text{Rust}} \triangleq \text{“Rust term } t \text{ from state } \sigma \text{ satisfies } \varphi\text{”}$

$\text{wp}(t, \varphi) \triangleq \text{weakest precondition on } \sigma \text{ s.t. } [t, \sigma \Vdash_L \varphi]_{\text{Rust}}$

Temporal formulas

- $\varphi ::= \text{now } P \mid \text{AG } \varphi \mid \varphi \text{ AU } \varphi' \mid \varphi \wedge \varphi'$
- Sugar: $\text{AF } \varphi' = \top \text{ AU } \varphi', \quad \text{AX } \varphi' = \top \text{ AN } \varphi'$
- done Q : termination postcondition; now P : intermediate state predicate

Inference Rules

Assignment

Rust: $*x = e$

$$\frac{\sigma' = \sigma[x \mapsto v] \quad \varphi(\sigma') \quad \varphi'(\sigma') \vee (\text{continue checking})}{[*x = e, \sigma \Vdash_L \varphi \textbf{AU} \varphi']_{\text{Rust}}} \text{ASSIGN}_{\text{RUST}} \text{AU}_L$$

Analogous to **Assign** in Imp: $\llbracket s \leftarrow x \rrbracket_s$ updates the state map.

Sequential Composition (let-binding)

Rust: `let x = t; u`

$$\frac{\begin{array}{c} [t, \sigma \Vdash_R \varphi \text{ AU AX done } \mathcal{R}]_{\text{Rust}} \\ \forall x, \sigma', \mathcal{R} \times \sigma' \rightarrow [u[x], \sigma' \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}} \end{array}}{[\text{let } x = t; u, \sigma \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}}} \text{LET}_{\text{RUST}} \text{AU}_L$$

Direct lift of **BindAU**_L / **SeqS**AU_L from TIDL.

Rust: `if c {t} else {u}`

$$\frac{\begin{array}{l} [c, \sigma \Vdash_R \varphi \text{ AU done} = (b, \sigma')]_{\text{Rust}} \\ \left\{ \begin{array}{ll} [t, \sigma' \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}}, & \text{if } b \\ [u, \sigma' \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}}, & \text{otherwise} \end{array} \right. \end{array}}{[\text{if } c \{t\} \text{ else } \{u\}, \sigma \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}}} \text{IF}_{\text{RUST}}\text{AU}_L$$

Evaluating c may take multiple steps; φ must hold along the path.

While Loop (AU – terminating)

Rust: `while c {t}` **with** invariant $\mathcal{R}$, decreases f

$$\begin{array}{c}
 \mathcal{R} \sigma \\
 \forall \sigma, \mathcal{R} \sigma \rightarrow [c, \sigma \Vdash_R \varphi \text{ AU done} = (b, \sigma')]_{\text{Rust}} \\
 \wedge \left\{ \begin{array}{l} [t, \sigma' \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}} \vee \\ [t, \sigma' \Vdash_R \varphi \text{ AU AF done } (\lambda \sigma''. \mathcal{R} \sigma'' \wedge f \sigma'' < f \sigma)]_{\text{Rust}}, \text{ if } b \\ \varphi'(\sigma'), \text{ otherwise} \end{array} \right. \\
 \hline
 [\text{while } c \{t\}, \sigma \Vdash_L \varphi \text{ AU } \varphi']_{\text{Rust}} \quad \text{WHILE}_{\text{RUST}} \text{AU}
 \end{array}$$

Loop (AG – infinite)

Rust: `loop {t}` **with** invariant $\mathcal{R}$ (**no** decreases)

$$\frac{\begin{array}{c} \mathcal{R} \sigma \\ \forall \sigma, \mathcal{R} \sigma \rightarrow \\ [\text{loop } \{t\}, \sigma \Vdash_L \varphi]_{\text{Rust}} \wedge \\ [t, \sigma \Vdash_R \varphi \text{ AU AF done } (\lambda \sigma' \Rightarrow \mathcal{R} \sigma')]_{\text{Rust}} \end{array}}{[\text{loop } \{t\}, \sigma \Vdash_L \text{AG } \varphi]_{\text{Rust}}} \text{LOOP}_{\text{RUST}} \text{AG}$$

Coinductive hypothesis $[\text{loop } \{t\}, \sigma \Vdash_L \varphi]_{\text{Rust}}$ enables proving φ holds at loop re-entry after each iteration.

Function Call (Bind Rule)

Rust: `let x = f(); k x`

$$\frac{\begin{array}{c} f \text{ ensures AF done } \mathcal{R} \\ \forall x, \sigma', \mathcal{R} \times \sigma' \rightarrow [k \ x, \ \sigma' \Vdash_L \ \varphi \text{ AU } \varphi']_{\text{Rust}} \end{array}}{[\text{let } x = f(); k \ x, \ \sigma \Vdash_L \ \varphi \text{ AU } \varphi']_{\text{Rust}}} \text{CALL}_{\text{RUST}} \text{AU}_L$$

Contract-based transparent checking (Vera extension)

If f has temporal ensures Ψ , additionally assert $\Psi \implies \Phi$ where Φ is the caller's temporal obligation.

Async Bind (Lazy Future Creation)

Rust: `let p = async f(); k p`

$$\frac{\text{No state change (lazy future)} \quad [k \ p, \ \sigma \Vdash_L \ \Phi]_{\text{Rust}}}{[\text{let } p = \text{async } f(); \ k \ p, \ \sigma \Vdash_L \ \Phi]_{\text{Rust}}} \text{ASYNCBIND}_{\text{RUST}}$$

Rust futures are lazy: `async fn()` returns a handle, body runs at `.await`.

Await (AU/AF – terminating callee)

Rust: `let x = p.await; k x`

$$\frac{\begin{array}{l} P[p] \text{ ensures } \varphi' \text{ AU AX done } \mathcal{R} \\ \varphi' \implies \varphi \text{ (transparent)} \\ \forall x, \sigma', \mathcal{R} \ x \ \sigma' \rightarrow [k \ x, \ \sigma' \Vdash_L \ \varphi \text{ AU } \varphi'']_{\text{Rust}} \end{array}}{[\text{let } x = p.\text{await}; \ k \ x, \ \sigma \Vdash_L \ \varphi \text{ AU } \varphi'']_{\text{Rust}}} \text{AWAIT}_{\text{RUST}} \text{AU}_L$$

$P[p]$: async process behind future p . Callee's path φ' must imply caller's φ (transparent await).

Await (AG – diverging callee)

Rust: `p.await` where $P[p]$ ensures AG ψ

$$\frac{\begin{array}{c} P[p] \text{ ensures AG } \psi \\ \psi \implies \varphi \end{array}}{[p.\text{await}, \sigma \Vdash_L \text{AG } \varphi]_{\text{Rust}}} \text{AWAIT}_{\text{RUST}} \text{AG}$$

Callee runs forever $\rightarrow$ await is a *divergence point*. Caller's AG φ discharged if $\psi \implies \varphi$.

Weakest Preconditions

$$\text{wp}(*x = e, \varphi \text{ AU } \varphi') = \varphi[\sigma'/\sigma] \wedge (\varphi'[\sigma'/\sigma] \vee \top)$$

where $\sigma' = \sigma[x \mapsto \llbracket e \rrbracket_\sigma]$.

wp: Let-binding (Sequential Composition)

$$\begin{aligned} \text{wp}(\text{let } x = t; u, \varphi \text{ AU } \varphi') &= \text{wp}(t, \varphi \text{ AU } \text{AX done } \mathcal{R}) \\ &\quad \wedge \forall x, \sigma'. \mathcal{R} \times \sigma' \rightarrow \text{wp}(u[x], \varphi \text{ AU } \varphi') \end{aligned}$$

$$\begin{aligned} \text{wp}(\text{if } c \{t\} \text{ else } \{u\}, \varphi \text{ AU } \varphi') &= \text{wp}(c, \varphi \text{ AU done} = (b, \sigma')) \\ &\wedge \begin{cases} \text{wp}(t, \varphi \text{ AU } \varphi')[\sigma'/\sigma], & \text{if } b \\ \text{wp}(u, \varphi \text{ AU } \varphi')[\sigma'/\sigma], & \text{otherwise} \end{cases} \end{aligned}$$

wp: While Loop (AU)

$$\begin{aligned} \text{wp}(\text{while } c \{t\}, \varphi \text{ AU } \varphi') &= \mathcal{R} \sigma \wedge \forall \sigma, \mathcal{R} \sigma \rightarrow \\ &\quad \text{wp}(c, \varphi \text{ AU done} = (b, \sigma')) \wedge \\ &\quad \left\{ \begin{array}{l} \text{wp}(t, \varphi \text{ AU } \varphi')[\sigma'/\sigma] \vee \\ \quad \text{wp}(t, \varphi \text{ AU AF done } (\lambda \sigma''. \mathcal{R} \sigma'' \wedge f \sigma'' < f \sigma)) \\ \quad [\sigma'/\sigma], \\ \varphi'(\sigma'), \end{array} \right. \end{aligned} \quad \begin{array}{l} \text{if } b \\ \text{o.w.} \end{array}$$

wp: Loop (AG)

$$\begin{aligned} \text{wp}(\text{loop } \{t\}, \text{AG } \varphi) = & \mathcal{R} \sigma \wedge \forall \sigma, \mathcal{R} \sigma \rightarrow \\ & [\text{loop } \{t\}, \sigma \Vdash_L \varphi]_{\text{Rust}} \wedge \\ & \text{wp}(t, \varphi \text{ AU } \text{AF done } (\lambda \sigma'. \mathcal{R} \sigma')) \end{aligned}$$

Coinductive: $[\text{loop } \{t\}, \sigma \Vdash_L \varphi]_{\text{Rust}}$ is the coinductive hypothesis.

wp: Function Call

$$\text{wp}(\text{let } x = f(); k \ x, \ \varphi \text{ AU } \varphi') = \forall x, \sigma'. \mathcal{R}_f \ x \ \sigma' \rightarrow \text{wp}(k \ x, \ \varphi \text{ AU } \varphi')$$

where $\mathcal{R}_f$ is f 's postcondition from ensures AF done $\mathcal{R}_f$.

Transparent: If f has temporal ensures Ψ , additionally $\Psi \implies \Phi$.

wp: Async Bind

$$\text{wp}(\text{let } p = \text{async } f(); k \ p, \Phi) = \text{wp}(k \ p, \Phi)$$

No-op: lazy future creation does not change state.

wp: Await (AU)

$$\begin{aligned} \text{wp}(\text{let } x = p.\text{await}; k\ x, \varphi \text{ AU } \varphi'') &= \varphi' \implies \varphi \\ &\wedge \forall x, \sigma'. \mathcal{R}\ x\ \sigma' \rightarrow \text{wp}(k\ x, \varphi \text{ AU } \varphi'') \end{aligned}$$

where $P[p]$ ensures $\varphi' \text{ AU } \text{AX done } \mathcal{R}$.

wp: Await (AG)

$$\text{wp}(p.\text{await}, \text{AG } \varphi) = \psi \implies \varphi$$

where $P[p]$ ensures $\text{AG } \psi$.

Callee diverges $\rightarrow$ continuation is dead code $\rightarrow$ no further wp.

Summary

Complete Rule Set

Rust Construct	TICL Rule	Imp Analog
<code>*x = e</code>	State update, check φ	Assign
<code>let x = t; u</code>	BindAU _L	Seq _S AU _L
<code>if c {t} else {u}</code>	Case split	If _S AU _L
<code>while c {t} + decreases</code>	IterAU _L	While _S AU _L
<code>loop {t} (no exit)</code>	IterAG	While _S AG
<code>f()</code>	Bind + transparent	BindAU _L
<code>let p = async f(); k</code>	No-op (lazy)	–
<code>p.await</code> (AU callee)	Bind + implication	BindAU _L
<code>p.await</code> (AG callee)	Divergence	–

Key extension over Imp: contract-based transparent temporal checking at function calls and `.await` boundaries.